

INFORMATION FOR FIRST AND SECOND RESPONDERS

EMERGENCY RESPONSE GUIDE FOR VEHICLES

**SUBARU
· HYBRID ELECTRIC VEHICLES**



V8440BE

Ver.02

2025.10

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Foreword

- This document describes cautionary matters to be observed when performing rescue work on a Subaru vehicle.
- In order to perform work safely, please thoroughly read this document and understand the structure and operations of Subaru vehicles before carrying out work.
- The illustrations featured in this document are representative examples. For vehicle-specific information such as parts and locations for each vehicle, refer to the RS (Rescue Sheet) for each vehicle.

■ Indications on safety

WARNING



Describes matters that, if not followed, may cause death or severe injuries.

CAUTION



Describes matters that, if not followed, may cause injuries or accidents.

NOTE

Describes matters that should be followed and useful tips.

0. Rescue Sheets

For North America

- In this chapter, you will find the names of all Subaru vehicles sold in North America, which are:
 - Hybrid Electric Vehicles
- The rescue sheets of all these vehicles are available on the website:
 - <https://www.nhtsa.gov/ERG>
- The models concerned by this ERG are:
 - Hybrid family:
 - Crosstrek Hybrid 2026MY-
 - Forester Hybrid 2025MY-

1. Identification / Recognition

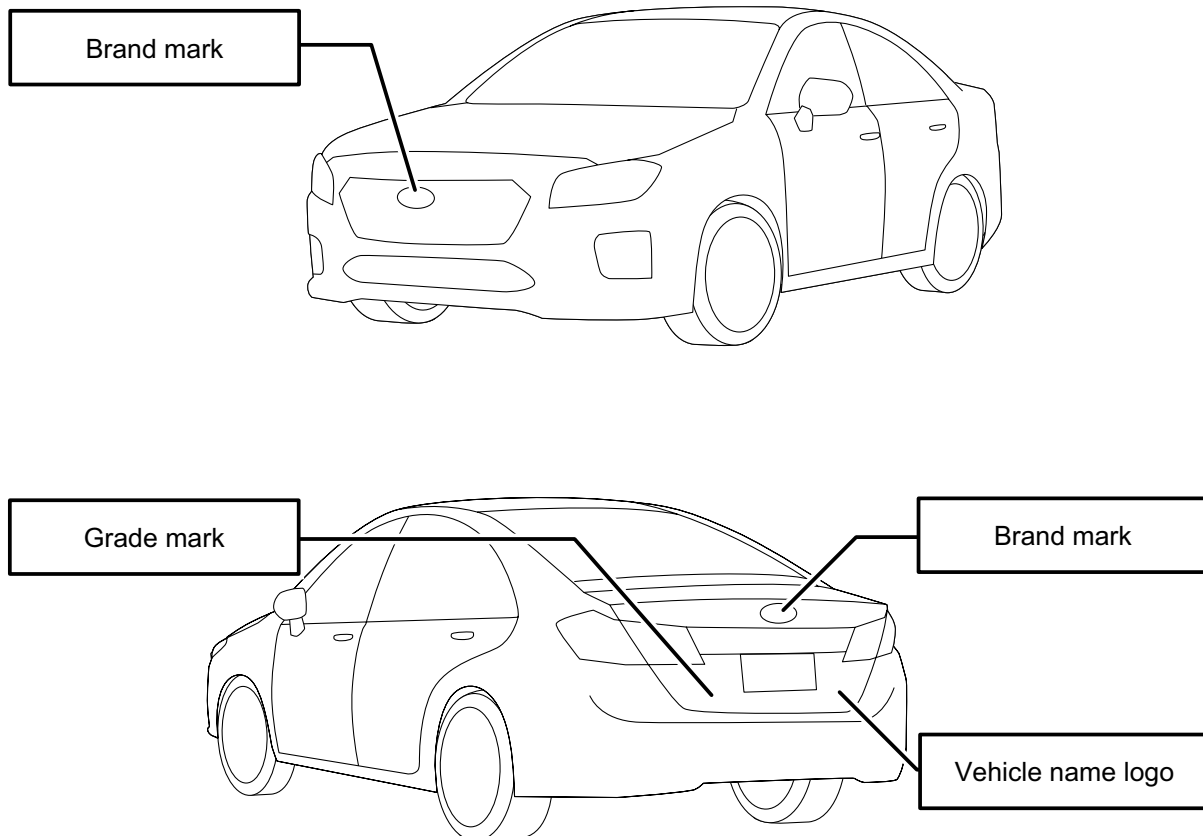
Appearance and logo

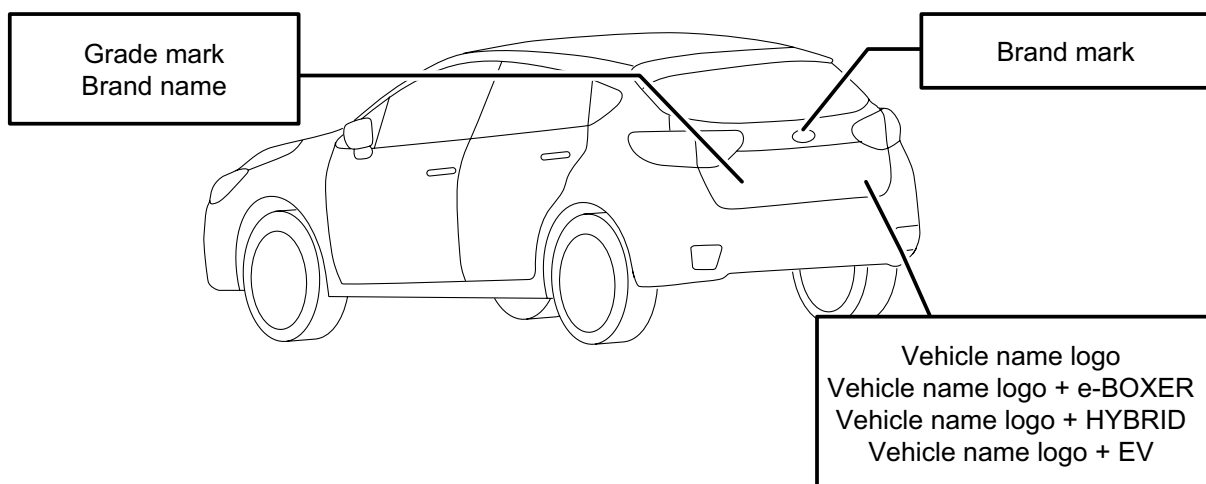
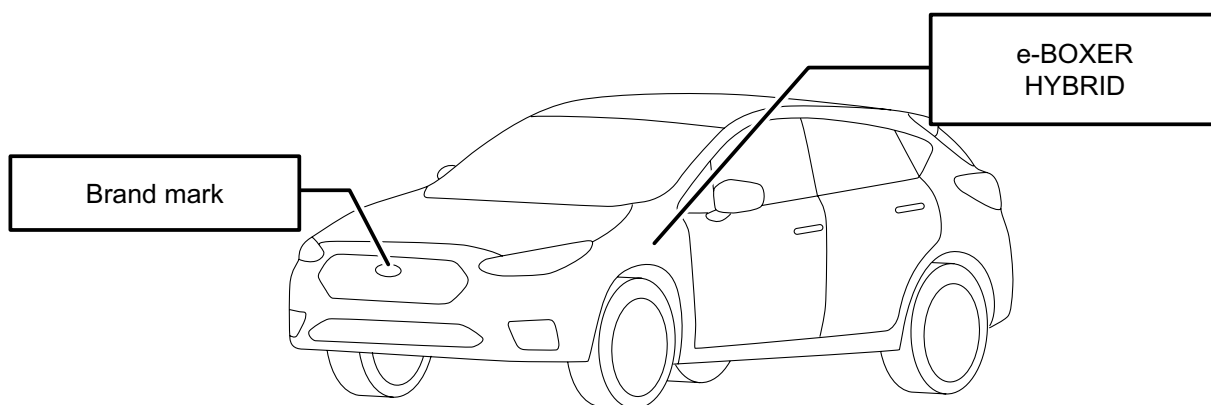
- Identify the model based on the properties of the outer appearance and information such as the body logo marks.
- Logo marks include the brand mark, marks that indicate the vehicle name and grade, and marks that indicate hybrid electric vehicles.
- Logo marks are attached to locations such as the trunk or rear gate, front grille, and front fender.

CAUTION



Lack of engine noise does not mean vehicle is off: silent movement or instant restart capability exists until vehicle is fully shut down. Wear appropriate PPE.



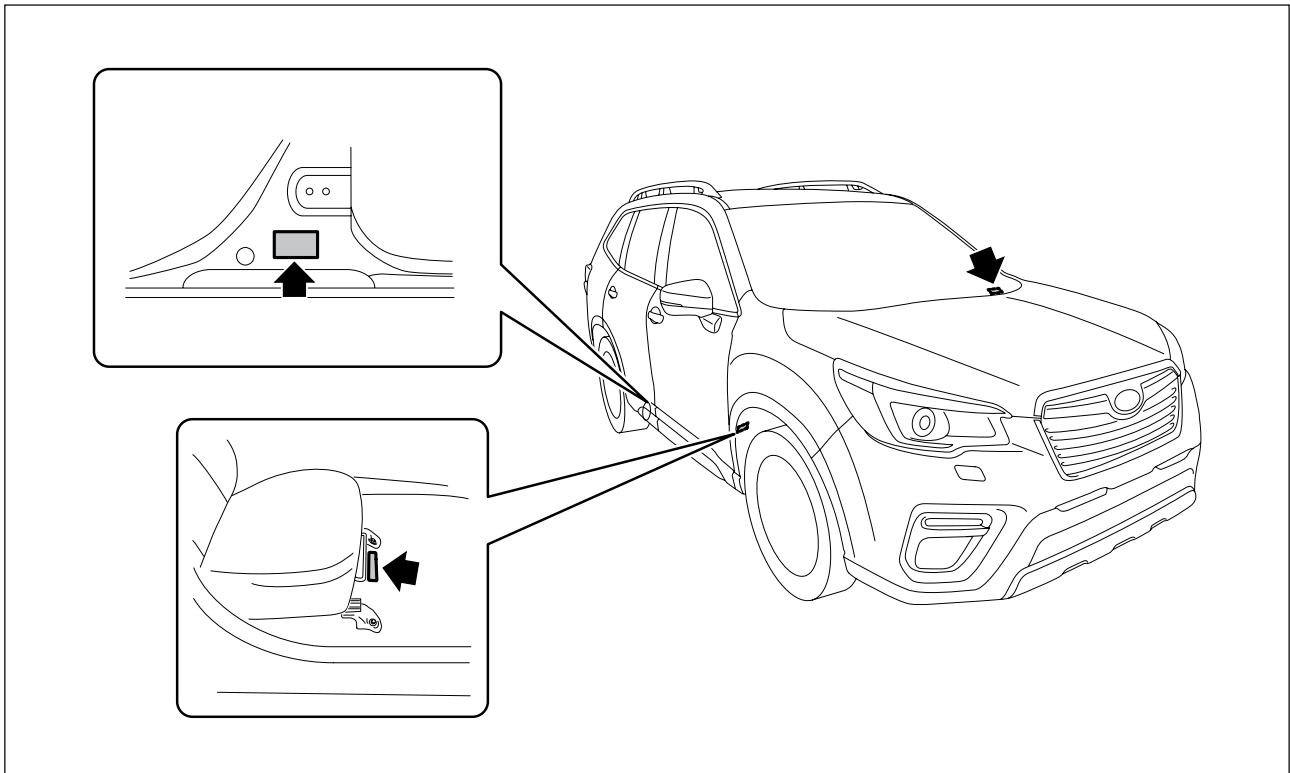


NOTE

For information about the logo marks location on each vehicle, refer to the RS (Rescue Sheet) for each vehicle.

Vehicle Identification Number (VIN)

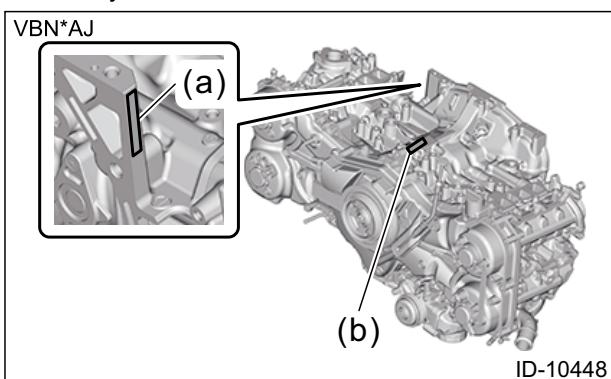
- The VIN is written on the windshield glass, the passenger's seat floor and B pillar.
- The characters in the VIN before the hyphen (for "YYYXXXYYYYYYYYYY," the "XXX" part) can be used to identify the model.



Engine type

- The engine type is written on the upper side of the crankcase.
- The character in the fifth digit of the engine type (for "YYYYXYYYYY," the "X" part) can be used to identify a hybrid electric vehicle.

W : Hybrid electric vehicle

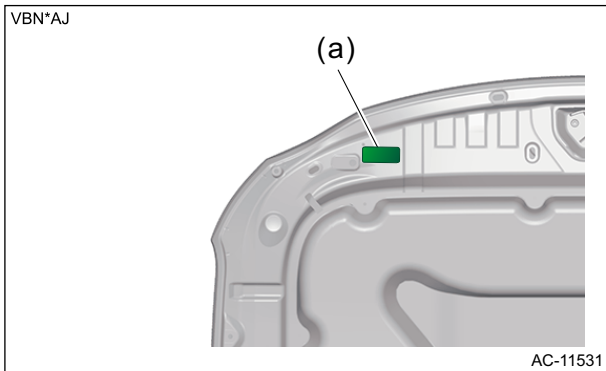
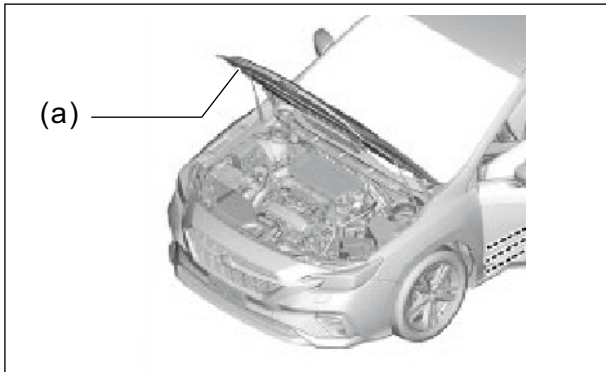


(a) Engine serial number

(b) Engine type (crankcase upper side)

A/C system refrigerant

The type of refrigerant used is written on each vehicle.



(a) Label

Battery equipped location

For information about the battery location on each vehicle, refer to the RS (Rescue Sheet) for each vehicle.

2. Immobilization / Stabilisation / Lifting

Completely immobilize the vehicle

WARNING



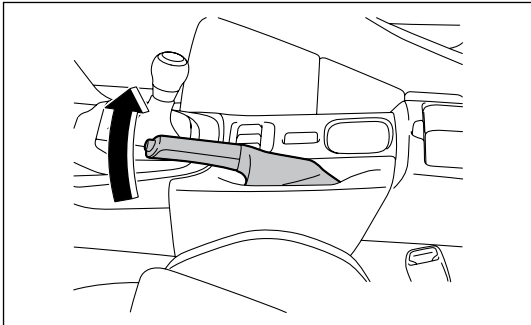
In cases where the relevant parties leave a hybrid electric vehicle (HEV) or electric vehicle (EV), there is a risk that other persons may touch the vehicle without knowing and suffer severe injuries or death due to electrical shock. In such a case, display a sign indicating "HIGH VOLTAGE. DO NOT TOUCH DURING OPERATION" to alert other people to the risk. (Make a photocopy of P36 for use.)

1. Set wheel stoppers and apply the parking brake.

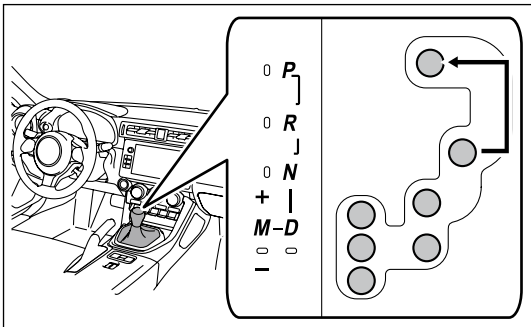
There are the following methods for applying the parking brake. Perform the operation that corresponds to the specifications.

■ Manual parking brake

- (1) To engage the parking brake, depress the brake pedal firmly and hold it down while you pull the parking brake lever up fully.



- (2) Shift the select lever to the Park position. (Automatic transmission model)



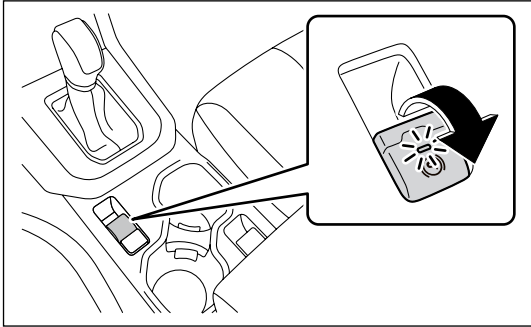
■ Electronic parking brake

NOTE

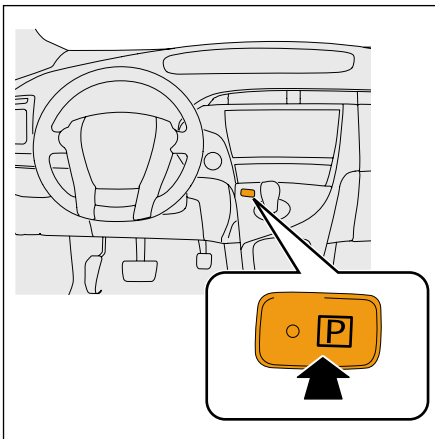
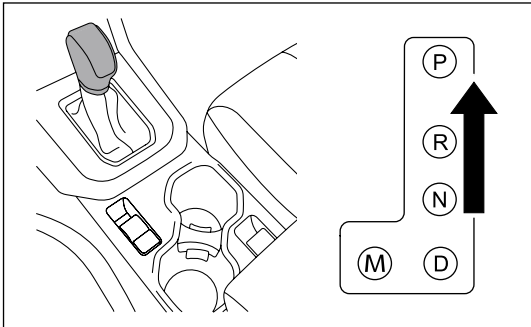
This cannot be operated if the battery (12 V) is not connected.

Perform this work before disconnecting the terminal of the battery (12 V).

(1) Chock wheels and engage the parking brake.



(2) Shift the select lever to the Park position, or press the Park position switch.



2. In order to perform rescue activities, take preliminary measures, such as opening the door glass, unlocking the doors, and opening the back door, as necessary.

NOTE

The following systems cannot be operated if the battery is not connected.

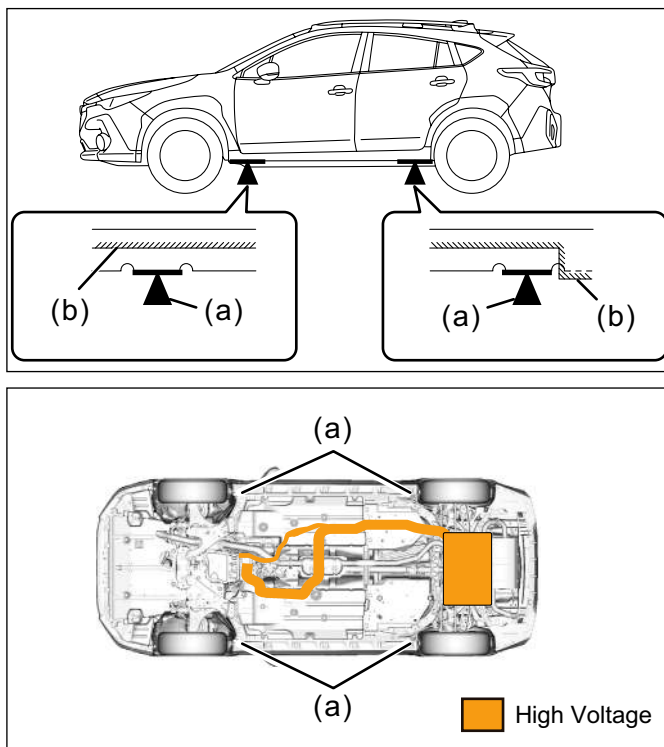
- Power door lock
- Power window
- Rear gate opener
- Electric parking brake
- Power seat controls
- Trunk opener
- Electro shiftmatic

Lifting point

Stabilize Vehicle

Crib at 4 points directly under the front and side sills.

Do not place cribbing under the orange high voltage cables, exhaust system, or fuel system.



(a) Cribbing Points

(b) Side Sill Garnish

NOTE

For information about the high voltage parts location on each vehicle, refer to the RS (Rescue Sheet) for each vehicle.

Preferred vehicle specific stabilisation points

To prevent substances such as gasoline and oil from leaking, we recommend moving the vehicle to a flat location where there is no fire or flammable material nearby before beginning work.

Checking the power status

- In order to safely perform rescue work, it is necessary to completely shut off the vehicle and cut off the power of the fuel pump, SRS airbag, high voltage battery, and the like before beginning work.
- Check the vehicle status, and if even one of the following items applies, the vehicle may be on and operational.

The engine is running.

The ignition key is in "ACC," "ON," or "START."

The meter is illuminated.

The air conditioning is on.

The wipers are operating.

The navigation system or display is on.

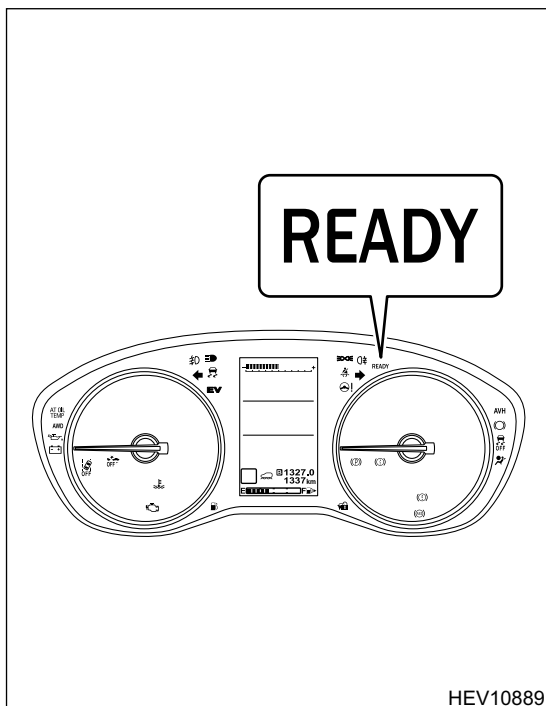
The charging cable is connected.

NOTE

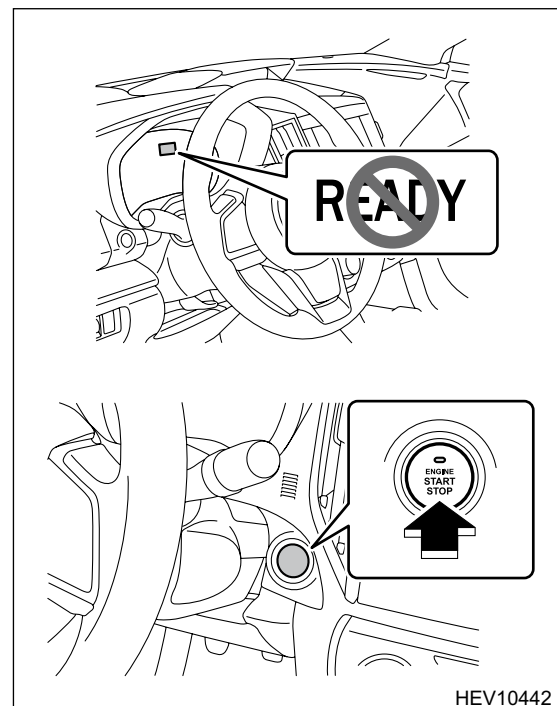
Before shutting off the vehicle and disconnecting the 12 V battery, if necessary, lower the windows, unlock the doors, open the rear gate, and move the power seats as required. Once the 12 V battery is disconnected, power controls will not operate.

1. Confirm the status of the READY indicator light in the combination meter.
2. If the READY indicator light is illuminated, the vehicle is on and operational. Shut off the vehicle by pushing the push-button ignition switch once.

▼ Hybrid READY Indicator Light



▼ Shut Off Vehicle (Hybrid READY Indicator Light-OFF)



The illustration above is a typical example. For some models, the combination meter may be slightly different than that shown in the illustration.

3. Disable direct hazards / Safety regulations

WARNING



- The high voltage system maintains voltage for as long as approximately 10 minutes after disabling the vehicle. If the high voltage system is not shut off, the high voltage may cause severe burns or severe injuries due to electrical shocks, and in the worst case, can cause death.
- In order to avoid severe burns or severe injuries or death due to electrical shock, do not touch the orange high voltage cables or high voltage parts (such as the high voltage battery). When it is necessary or possible to touch them, make sure to wear insulated gloves and insulate the orange high voltage cables and high voltage parts using insulating tape.
- Be sure to check the READY indicator light in the meter and determine if the high voltage system is active or shut off. If the READY indicator light is off, then the high voltage system is shut off.
- When the charging cable is connected, the high voltage system may be active even if the READY indicator light is off. Remove the charging cable before performing work.

Checking the power status

Refer to "Checking the power status" (Refer P11).

Disable Vehicle

■ Electric vehicle

WARNING



In order to avoid severe burns or severe injuries or death due to electrical shock, do not touch the orange high voltage cables or high voltage parts (such as the high voltage battery).

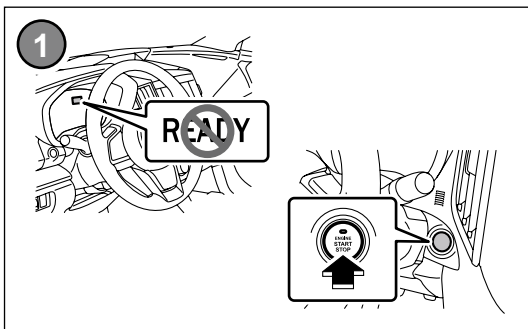
When it is necessary or possible to touch them, make sure to wear insulated gloves and insulate the orange high voltage cables and high voltage parts using insulating tape.

High-voltage circuit cutoff and 12V circuit disable procedure

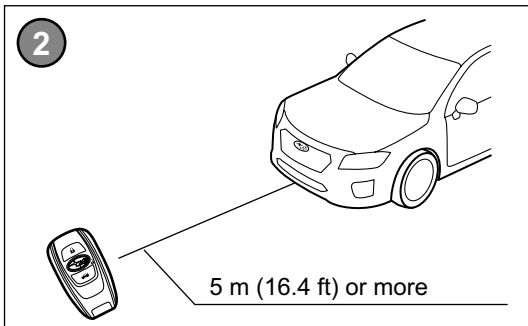
1. Turn the ignition switch to OFF.



(1) Check if the READY indicator light is off. If it is illuminated, press the push button ignition switch and turn the power off.



(2) Move the access key 5 m or more away.



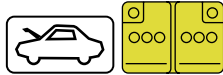
2. 12V circuit interruption

CAUTION

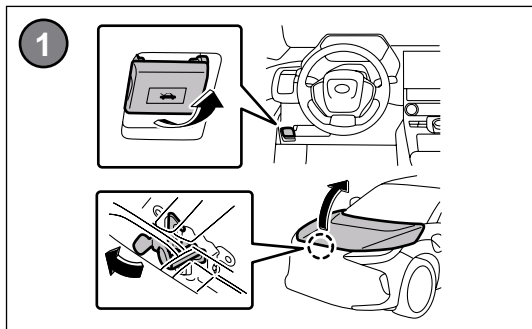


If the ground terminal of the 12 V battery cannot be removed, the 12 V circuit cannot be disabled. Be careful when working, as the SRS Airbag System may suddenly deploy due to a short circuit or if force is applied to the wiring harness, which may result in serious injury.

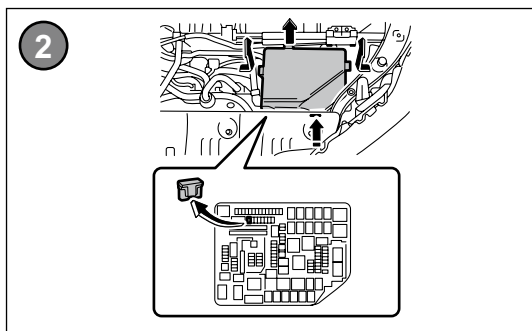
<Disconnect the ground terminal from battery sensor>



(1) Open the engine hood (bonnet). (Refer to P26)



(2) Remove the fuse.



NOTE

For information about the fuse location on each vehicle, refer to the RS (Rescue Sheet) for each vehicle.

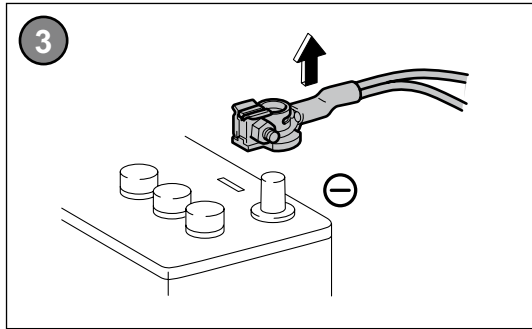
CAUTION



If you cannot find the said fuse or are having difficulty determining which fuse to remove, then remove fuses from the fuse box until all the following criteria apply.

- The meter is off.
- The air conditioning is not operating.
- The audio is not operating.
- The wipers are stopped.
- The navigation system and display are off.

(3) Disconnect the ground terminal from the battery.



CAUTION



- Cut off the electrical system to prevent the vehicle from restarting as well as electrical fires.
- The integrated capacitor power supply maintains a voltage of approximately 12 V between the positive and ground terminals of the integrated capacitor power supply for as long as approximately 10 minutes after the ground terminal has been cut from the 12 V battery and power has been cut off.

■ Hybrid electric vehicle

WARNING



In order to avoid severe burns or severe injuries or death due to electrical shock, do not touch the orange high voltage cables or high voltage parts (such as the high voltage battery).

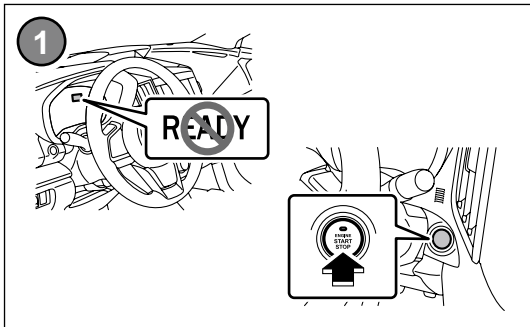
When it is necessary or possible to touch them, make sure to wear insulated gloves and insulate the orange high voltage cables and high voltage parts using insulating tape.

High-voltage circuit cutoff and 12 V circuit disable procedure

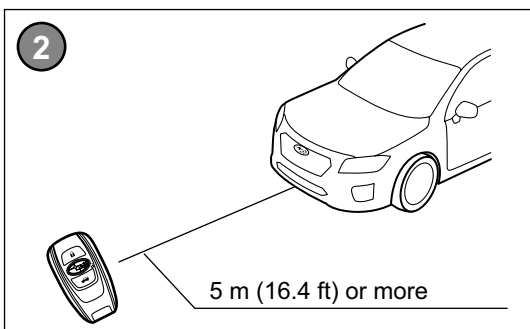
1. Turn the ignition switch to OFF.



(1) Check if the READY indicator light is off. If it is illuminated, press the push button ignition switch and turn the power off.



(2) Move the access key 5 m or more away.



2A. <Main disabling method>

Remove the service plug.

CAUTION

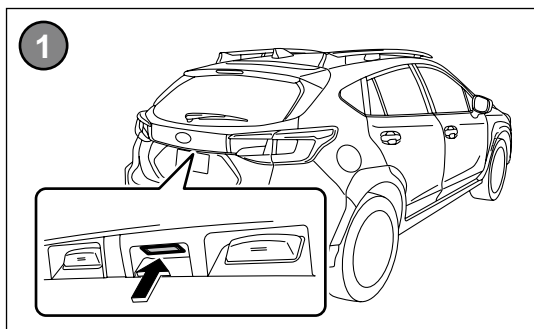


- If you cannot disable the high-voltage circuit, do not touch, disconnect, or open the high-voltage components or high-voltage battery.
- To prevent injury, wear personal protective equipment suitable for organic solvents including SCBA.

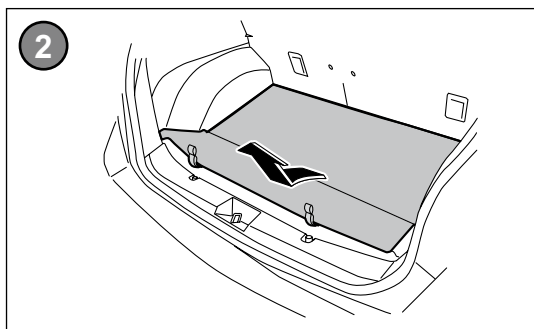


In case of a collision with airbag deployment and/or seat belt pretensioner activation, the hybrid system and fuel pump will be disabled. The restraint systems are still operational.

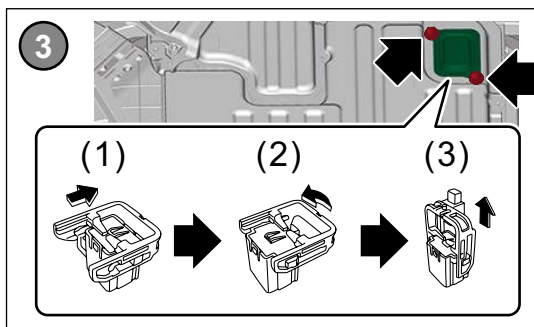
(1) Press the rear gate opener button and open the rear gate. (Refer to P27)



(2) Remove the rear floor mat.



(3) Remove the service disconnect plug cover and then remove the service disconnect plug.



CAUTION



After removing the service disconnect plug, the worker should carry it on their person to prevent others from accidentally replacing it while work is still being performed. Cover the connector on the vehicle side with insulating tape and secure the plug cover.

NOTE

For information about the service disconnect plug location on each vehicle, refer to the RS (Rescue Sheet) for each vehicle.

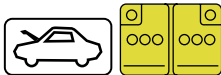
2B. <Alternative disabling method>

CAUTION

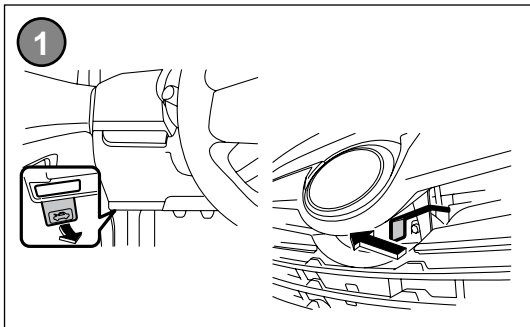


If the ground terminal of the 12 V battery cannot be removed, the 12 V circuit cannot be disabled. Be careful when working, as the SRS Airbag System may suddenly deploy due to a short circuit or if force is applied to the wiring harness, which may result in serious injury.

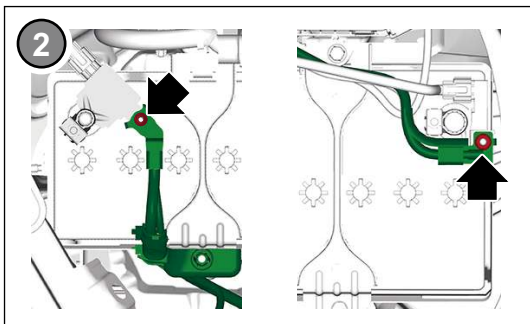
Disconnect the ground terminal from battery sensor



(1) Open the engine hood (bonnet). (Refer to P26)



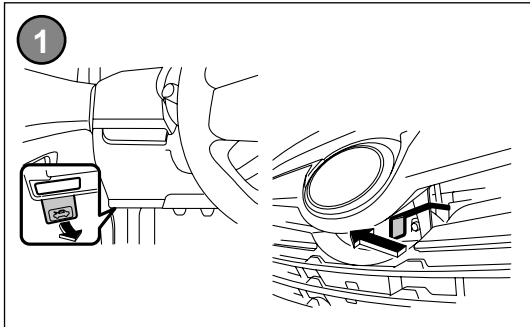
(2) Disconnect the ground terminal from the battery.



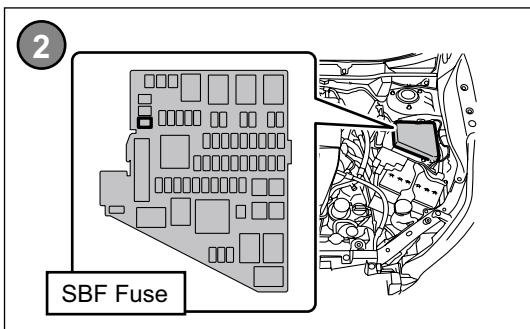
Alternatively, remove the FUSE



(1) Open the engine hood (bonnet). (Refer to P26)



(2) Remove the fuse.



NOTE

For information about the fuse location on each vehicle, refer to the RS (Rescue Sheet) for each vehicle.

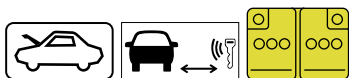
CAUTION



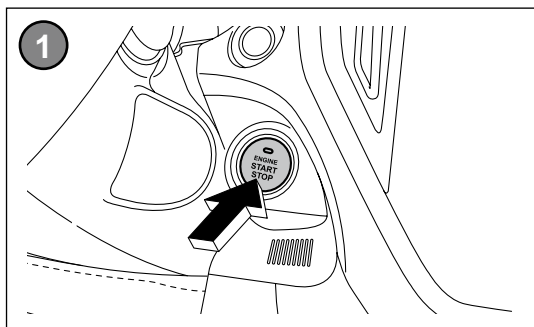
If you cannot find the said fuse or are having difficulty determining which fuse to remove, then remove fuses from the fuse box until all the following criteria apply.

- The meter is off.
- The air conditioning is not operating.
- The audio is not operating.
- The wipers are stopped.
- The navigation system and display are off.

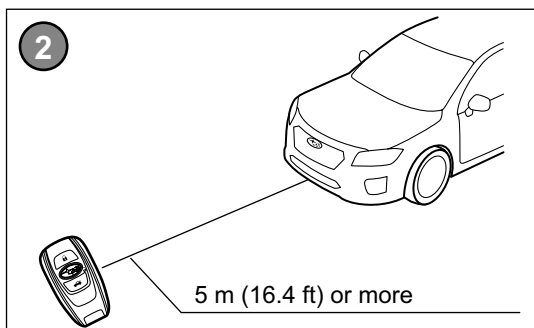
■ Gasoline vehicle



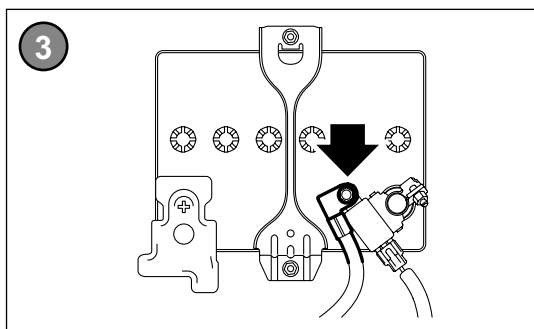
- (1) Check if the READY indicator light is off. If it is illuminated, press the push button ignition switch and turn the power off.



- (2) Move the access key 5 m or more away.

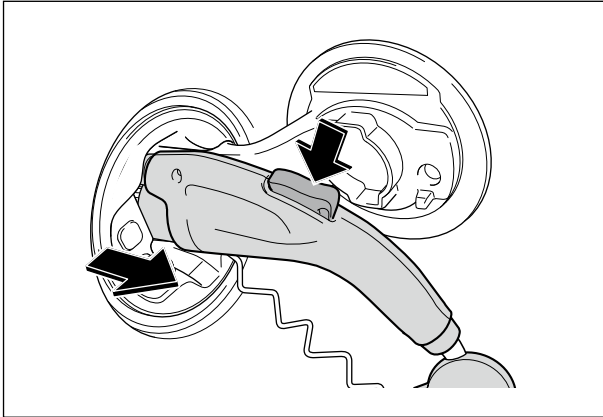


- (3) Disconnect the ground terminal from the battery.



Electric vehicle (EV)

- Electric vehicles are equipped with a function for connecting to external electricity to charge the high voltage battery.
- If the charging cable is connected to the vehicle's charging inlet, follow the procedure below to remove the charging cable and stop power supply.
 1. Turn off the main breaker for the external power supply to stop the power supply.
 2. While pressing the lock release button on the charging connector, pull it towards you and remove the charging cable from the charging inlet.

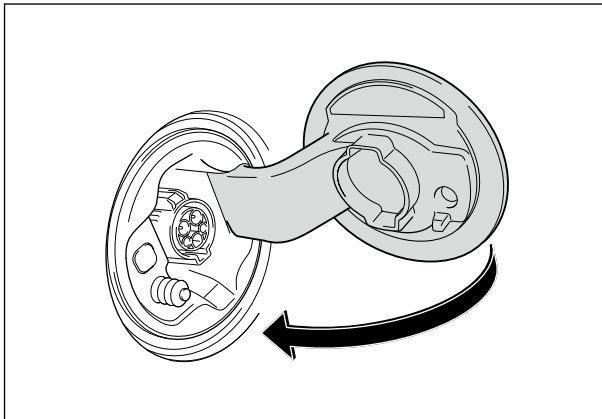


CAUTION

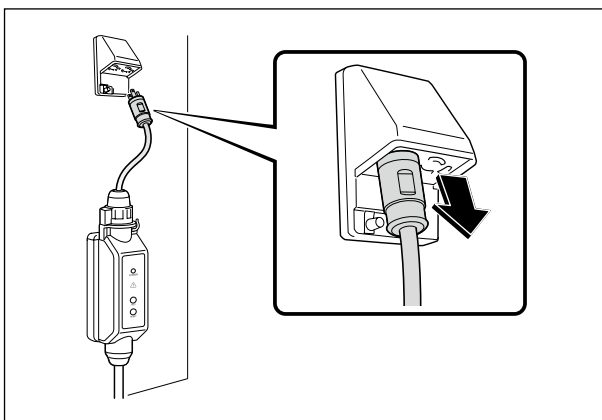


- Some electric vehicles have a function for locking the charging cable. The lock can be released by pressing the connector lock switch with the door unlocked.
- If the charging cable lock cannot be released during normal charging, either remove the power supply plug for the external power supply from the outlet, or turn off the main breaker of the external power supply.
- The charging cable cannot be released during quick charging. If charging does not stop even after performing the stop operation for quick charging, turn off the main breaker for the quick charging stand.

3. Close the charger port of the charging inlet.



4. Either remove the power supply plug for the external power supply from the outlet, or stop the external power supply function.



4. Access to the occupants

Side impact protection beam

The side impact protection beam is located inside the door panel.

High tensile strength steel panel

Some vehicles use a super high tensile strength steel panel (1.5 GPa class), which provides strength approximately 1.5 times that of the normal high tensile strength steel panel (less than 1 GPa class).

For information about the position where the high tensile strength steel panel is used on each vehicle, refer to the RS (Rescue Sheet) for each vehicle.

Glass

- Glass used in automobiles is generally split into two categories: laminated glass and tempered glass.
- Laminated glass is used mainly in the windshield glass.
- Tempered glass is used mainly in the door glass, roof glass, and rear gate (window) glass.
- Laminated glass and tempered glass can be identified by a display on the corner of the glass (laminated glass = "LAMILEX", tempered glass = "TEMPERLITE" or "THERLITE-T", "TEMPALEX").

LAMILEX

**TEMPERLITE
THERLITE-T
TEMPALEX**

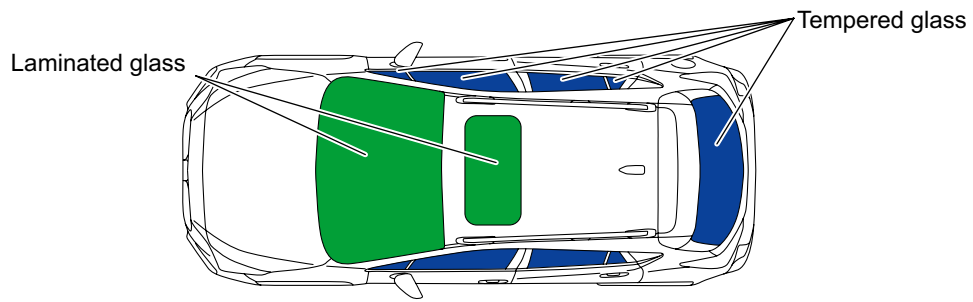
■ Laminated glass

Laminated glass is characterized by the layer of film laid between two layers of glass. This makes it resistant to penetration, and helps prevent shards from flying even if it is broken.

■ Tempered glass

Tempered glass is heated close to its softening temperature and then rapidly cooled to achieve three to five times the strength of normal glass. If it breaks, its shards become granular.

Glass (Used location)



NOTE

For information about locations where glass is used, refer to the RS (Rescue Sheet) for each vehicle.

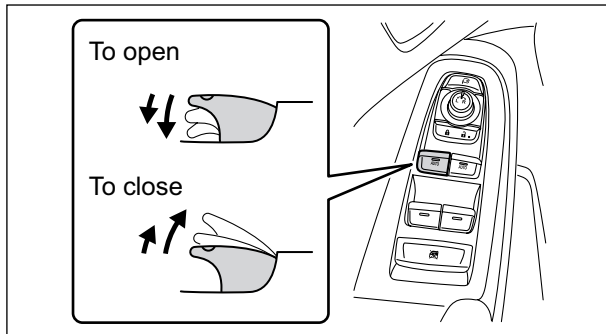
Power windows / Door locks

NOTE

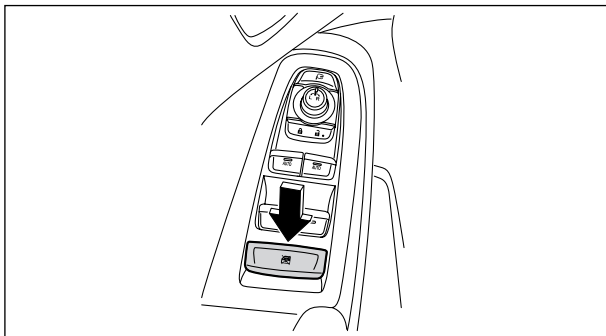
This cannot be operated if the battery (12 V) is not connected.

Perform this work before disconnecting the terminal of the battery (12 V).

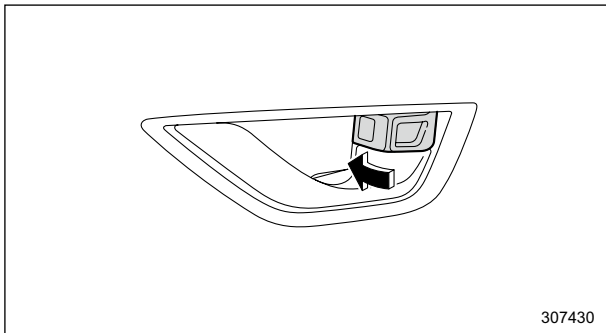
▼ Power window switch



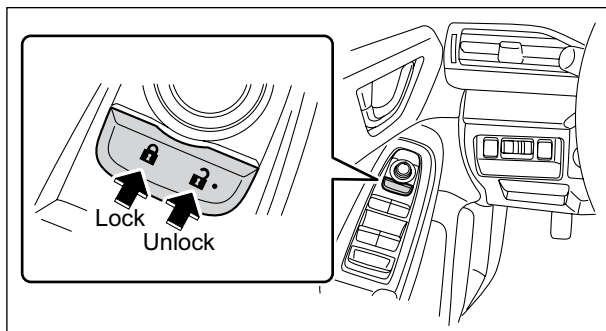
▼ Lock switch



▼ Door lock



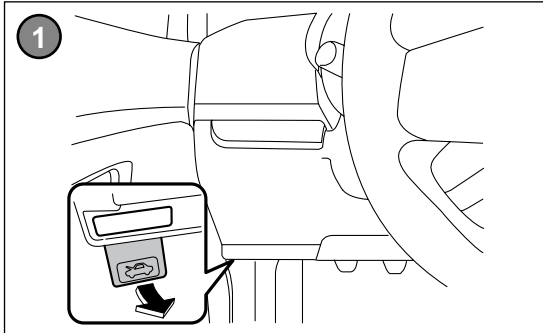
▼ Central door lock



Engine hood (Bonnet)



- (1) Pull the lever located on the driver's bottom right to raise the engine hood (bonnet) a bit.

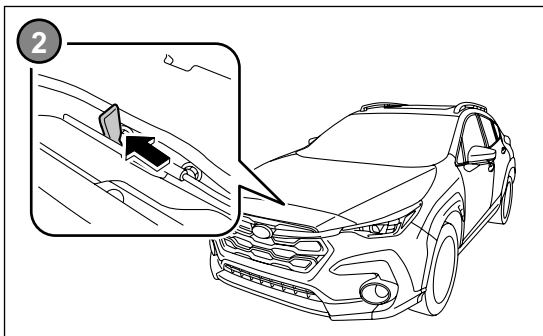


NOTE

For information about the lever location on each vehicle, refer to the RS (Rescue Sheet) for each vehicle.

- (2) Put your hand into the engine hood (bonnet) gap and press the lever in the central part of the front end to the left as you raise the engine hood (bonnet).

Insert the support pole into the engine hood (bonnet) hole and reliably secure the engine hood (bonnet).



NOTE

Some vehicles are equipped with dampers and do not have a support pole.

Rear gate



■ Method for opening the rear gate from the vehicle exterior

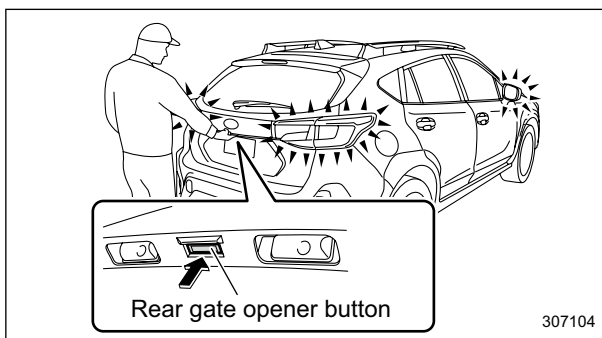
NOTE

This cannot be operated if the battery (12 V) is not connected.

Perform this work before disconnecting the terminal of the battery (12 V).

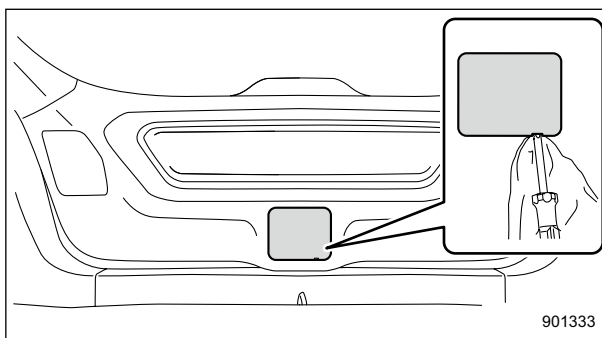
After releasing the door lock, press the rear gate opener button and lift the rear gate up a bit.

For the power rear gate, the rear gate automatically opens when pressing the rear gate opener button while holding the access key.



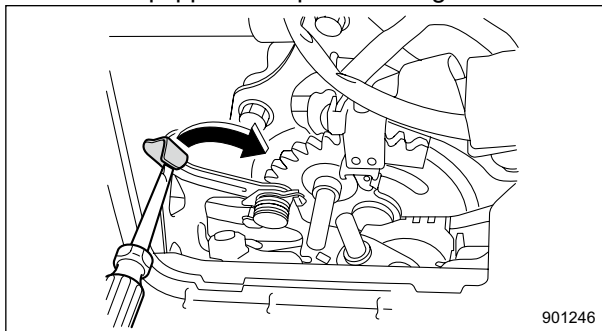
■ Method for opening the rear gate from the vehicle interior

Remove the trim cover of the rear gate.

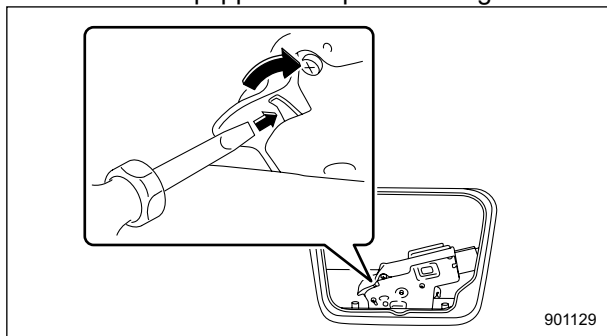


Use a flat tip screwdriver or the like and move the lever in the direction of the arrow, then press the rear gate and open it.

▼ Vehicle equipped with power rear gate



▼ Vehicle not equipped with power rear gate

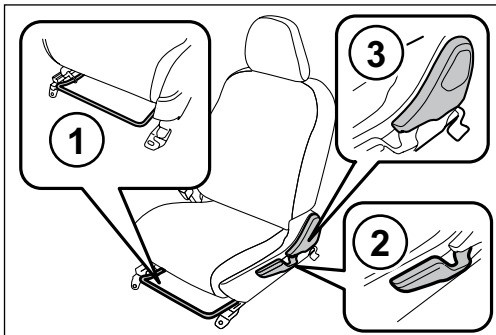


Front seat

There are manual seats, where the seat position is adjusted by operating levers and knobs, and power seats, where the seat position is adjusted by operating switches.

■ Manual seat

- (1)  Slide adjustment: Lift up the slide lever to adjust the seat position forward and backward.
- (2)  Lifter adjustment: Raise or lower the lifter lever to adjust the entire seat surface upward and downward.
- (3)  Reclining adjustment: Lift up the reclining lever to adjust the position of the backrest forward and backward.



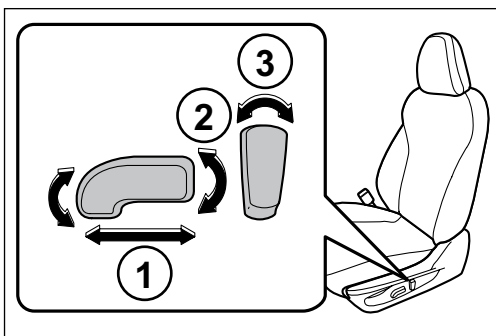
■ Power seat

NOTE

This cannot be operated if the battery (12 V) is not connected.

Perform this work before disconnecting the terminal of the battery (12 V).

- (1)  Slide adjustment: Use the seat switch slide adjustment to adjust the seat position forward and backward.
- (2)  Lifter adjustment: Use the seat switch lifter adjustment to adjust the entire seat surface upward and downward.
- (3)  Reclining adjustment: Use the seat switch reclining adjustment to adjust the position of the backrest forward and backward.

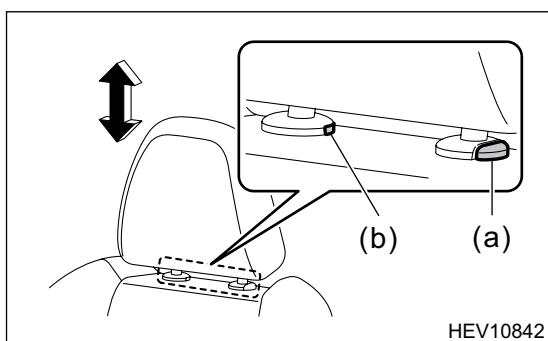


Headrest

- The headrest can be adjusted upward and downward and in the forward and backward inclination.
- To raise the headrest, lift up the headrest. To lower it, press and hold the release button as you press the headrest down.
- If removing the headrest,

■ Pattern A

1. Press and hold the lock release button as you lift the headrest up.
2. At the position where the headrest is locked, press the remove button and then lift the headrest up again to remove it.

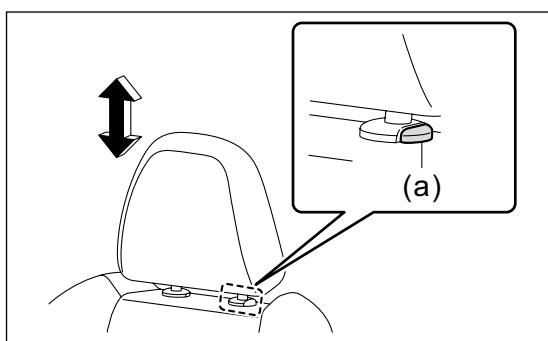


(a) Lock release button

(b) Removal button

■ Pattern B

1. Move the headrest to the uppermost position.
2. While pressing the lock release button, lift up the headrest and remove it.



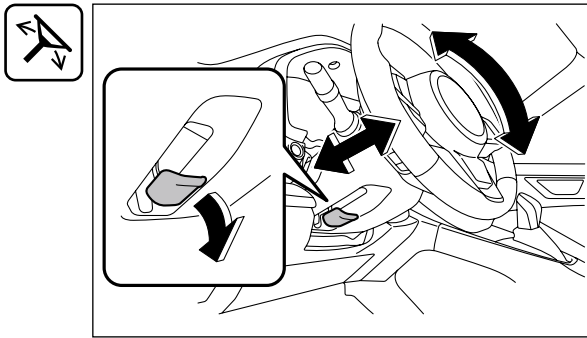
(a) Lock release button

Seat belt

- Push the button on the buckle enables the seat belt to be removed.
- If you cannot remove it even after pressing the release button, cut the belt.

Steering

Underneath the steering column, there is a lock lever used to release the lock of steering position.



Caution for when cutting the vehicle

WARNING



- In order to avoid severe burns or severe injuries or death due to electrical shock, do not touch the orange high voltage cables or high voltage parts (such as the high voltage battery). When it is necessary or possible to touch them, make sure to wear insulated gloves and insulate the orange high voltage cables and high voltage parts using insulating tape.
- As a fire or the like caused by sparks poses the risk of serious injury to rescuers and occupants, use a device such as a hydraulic cutter that does not create sparks to cut the vehicle.
- As the airbag system operates for 60 seconds after the power was turned off or the ground terminal of the 12 V battery was cut, check the time that has passed before carrying out the work.



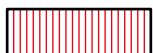
Location with risk of electrical shock due to high voltage.

Cannot be cut because there is a risk of electrical shock due to high voltage.



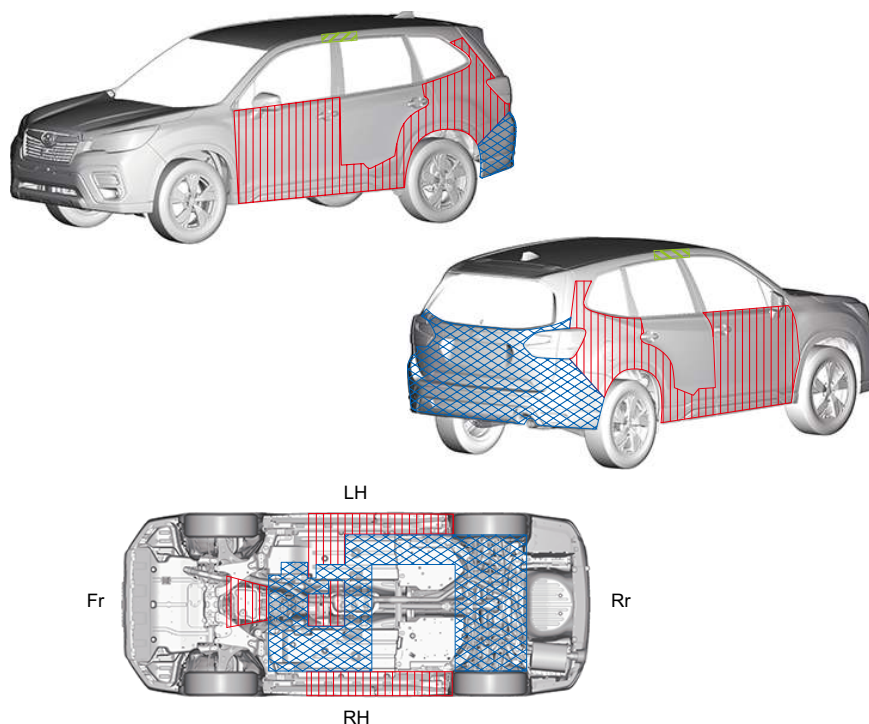
Location with risk of curtain airbag deploying.

Cannot be cut because there is a device that generates high-pressure gas used for deploying the curtain airbag. However, this can be cut if the curtain airbag is already deployed.



Location with risk of side airbag or curtain airbag deploying.

Cannot be cut because there is a risk of a wiring short circuit or impact causing the side airbag or curtain airbag to deploy. However, this can be cut if the side airbag or curtain airbag on the side to be cut is already deployed, or if 60 seconds or more has passed since the power was turned off or since the ground terminal of the 12 V battery was cut.



HEV10927

5. Stored energy / Liquids / Gases / Solids

Used liquid

For information about the liquids used in each vehicle, refer to the RS (Rescue Sheet) for each vehicle.

Battery

■ 12 V battery

- The 12 V battery supplies power to the computer that controls each system as well as devices such as the electric door locks, power windows, and power seats.
- In order to safely perform rescue work, it is necessary to completely shut off the vehicle through the vehicle stopping process (refer to P11). Before starting work, cut the ground terminal of the 12 V battery and cut off the electrical system to prevent the vehicle from restarting as well as electrical fires.
- Dilute sulfuric acid is used in the electrolyte of the 12 V battery.
- The 12 V battery is installed inside the engine or electric motor compartment.
- Under normal conditions of use, the battery does not present any risk of exposure to its content.
- For information about the 12 V battery location on each vehicle, refer to the RS (Rescue Sheet) for each vehicle.

WARNING



- As the hydrogen gas generated by the 12 V battery may ignite and explode, do not bring fire close to the 12 V battery or cause sparks near the 12 V battery.
- Dilute sulfuric acid is harmful to the human body and may cause inflammation if it is brought into contact with skin or the like. When it is necessary or possible to touch the electrolyte, wear appropriate protective equipment such as rubber gloves and protective goggles when performing work.

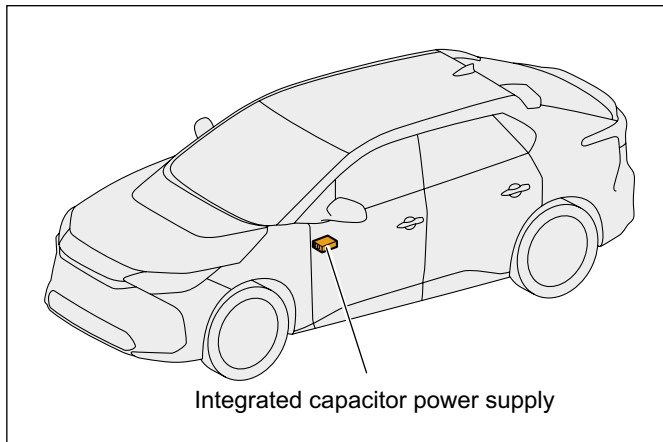
CAUTION



- When the 12 V battery is disabled using the vehicle stopping process (refer to P11), then the electrical auxiliary devices can no longer be operated. In order to facilitate rescue activities, take preliminary measures for the auxiliary devices, such as opening the door glass, unlocking the doors, and opening the back door, before cutting off the 12 V battery.
- The electrolyte includes substances that corrode the coated surface of the vehicle body. If it comes in contact with the vehicle body, there is the risk of changes such as the surface becoming cloudy.

■ Integrated capacitor power supply

- The integrated capacitor power supply is equipped inside the center console.
- Functions such as shift-by-wire and remote parking can be operated even after cutting off the battery.
- The internal capacitor is used to ensure a power supply.



CAUTION

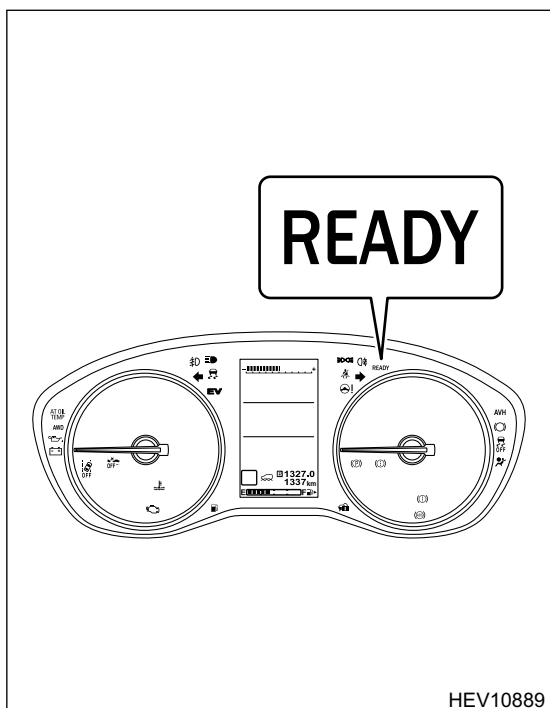


The integrated capacitor power supply maintains a voltage of approximately 12 V between the positive and ground terminals of the integrated capacitor power supply for as long as approximately 10 minutes after the ground terminal has been cut from the 12 V battery and power has been cut off.

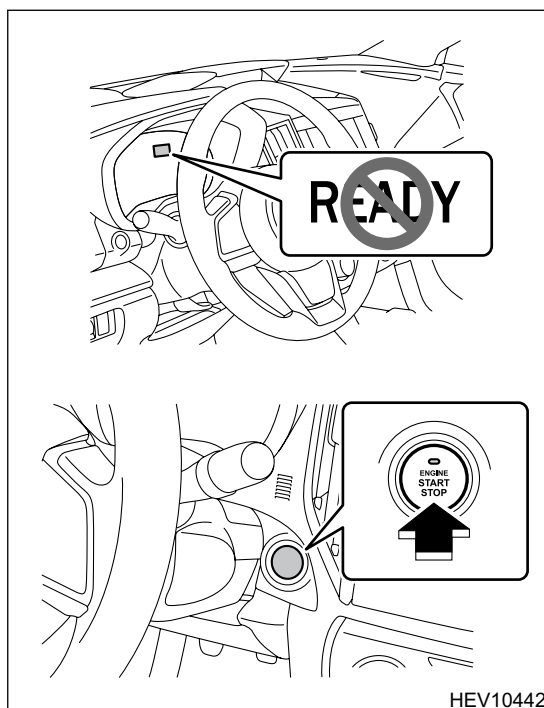
■ High voltage system

- Hybrid electric vehicles (HEV) and electric vehicles (EV) use a motor as the driving force, and use extremely high voltage to drive the motor. These vehicles are equipped with high voltage cables and high voltage parts such as the high voltage battery, inverter / converter, transmission / transaxle (motor) as high voltage devices.
- High voltage devices are equipped with a case, cover, or the like which indicates a high voltage device. Also, all the coverings for high voltage cables are colored in orange.
- The case / cover and device interior high voltage conductive parts for high voltages devices are insulated. Also, as both the high voltage devices and the vehicle body are insulated, touching the vehicle body will not cause an electrical shock when the vehicle is in a normal state.
- When the high voltage system is starting up, the READY indicator light in the meter illuminates.

▼ Hybrid READY Indicator Light



▼ Shut Off Vehicle
(Hybrid READY Indicator Light-OFF)



- The high voltage system shuts off when the ignition switch or power switch is set to OFF and the high voltage is cut off. Also, when a collision is detected (= when an airbag operates) or when the system detects electrical leakage on the high voltage system, the high voltage is automatically cut off. When the high voltage is cut off, the READY indicator light goes off.
- For detailed locations where the high voltage devices are equipped on each vehicle, refer to the RS (Rescue Sheet) for each vehicle.

WARNING



- The high voltage system maintains voltage for as long as approximately 10 minutes after disabling the vehicle (refer to P11). If the high voltage system is not shut off, the high voltage may cause severe burns or severe injuries due to electrical shocks, and in the worst case, can cause death.
- In order to avoid severe burns or severe injuries or death due to electrical shock, do not touch the orange high voltage cables or high voltage parts (such as the high voltage battery). When it is necessary or possible to touch the high voltage parts or high voltage cables, wear appropriate protective equipment such as insulated gloves.
- In cases where a vehicle is stored after responding to the accident or the like and the relevant parties leave the vehicle, there is a risk that other persons may touch the vehicle without knowing and suffer severe injuries or death due to electrical shock. In such a case, display a sign indicating "HIGH VOLTAGE. DO NOT TOUCH DURING OPERATION" to alert other people to the risk. (Make a photocopy of P36 for use.)

HIGH VOLTAGE CAUTION LABEL

_____ Person in charge:	
DURING OPERATION DO NOT TOUCH HIGH VOLTAGE. CAUTION:	
	CAUTION: HIGH VOLTAGE. DO NOT TOUCH DURING OPERATION
Person in charge: _____	
Copy this page and put it after folding on the roof of the vehicle in service.	

■ High voltage battery

The high voltage battery for driving the motor stores an extremely high voltage, and is equipped in the cargo area or underneath the vehicle.

Under normal conditions of use, the battery does not present any risk of exposure to its content.

WARNING



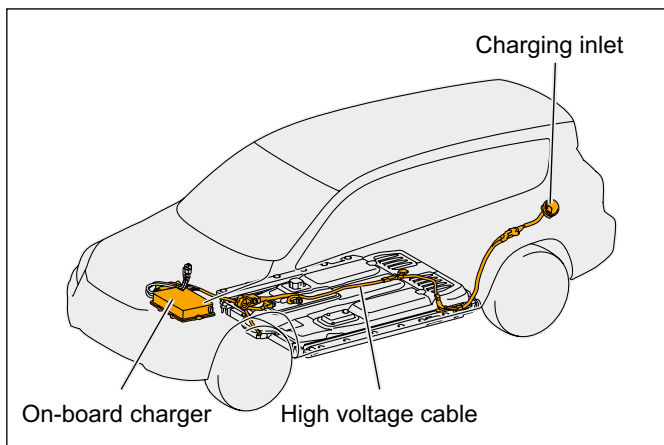
- The flammable organic electrolyte composed mainly of carbonic acid ester is harmful to the human body. If you touch the electrolyte, it may irritate your eyes, nose, throat, or skin. Also, if you touch the steam or smoke generated from leaked electrolyte or a burning battery, it may irritate your eyes, nose, throat, or skin. When it is necessary or possible to touch the electrolyte, wear appropriate protective equipment such as rubber gloves, protective goggles, a protective mask, and a self-contained breathing apparatus (SCBA) when performing work.
- If electrolyte leaked, keep it away from fire and keep good ventilation. Absorb the leaked electrolyte with a waste cloth or the like to collect in an enclosed container.
- The battery assembly cover should never be breached or removed under any circumstances, the high voltage may cause severe burns or severe injuries due to electrical shocks, and in the worst case, can cause death.

■ High voltage cables

- High voltage cables are all covered in orange, and are connected to the high voltage parts such as the high voltage battery, inverter / converter, and motor.
- High voltage cables are located in the engine compartment and vehicle center (center tunnel), or on the left or right side under the floor away from the rocker panel.

■ Plug-in charging system

- Electric vehicles are equipped with a charging system for charging the high voltage battery from an external source.
- The plug-in charging system is mainly comprised of the on-board charger and charging inlet.
- The on-board charger converts and boosts the AC voltage from the external power supply to DC high voltage, charging the high voltage battery.
- A charging inlet is equipped as a charging port for charging the high voltage battery from an external power supply, and a separate quick charging inlet is equipped for using quick charging stands (500 VDC).
- An orange high voltage cable is connected to the charging inlet, and a high voltage flows here during charging.



WARNING

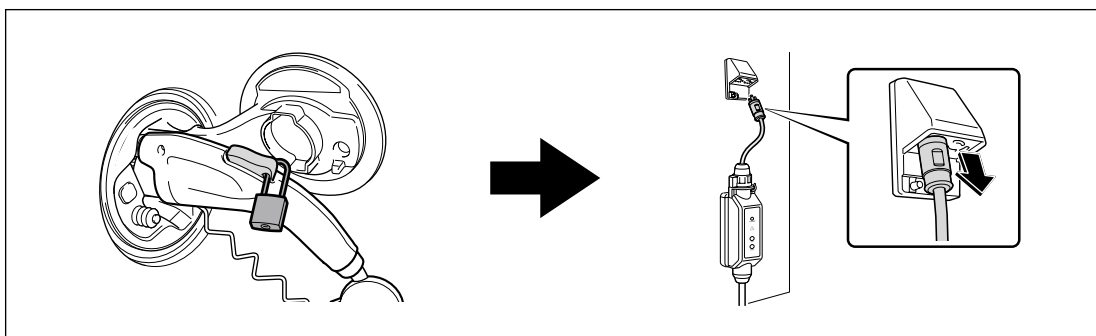


In order to avoid severe burns or severe injuries or death due to electrical shock, if the vehicle, external power supply, charging cable, or the like has been submerged, first turn off the main breaker of the external power supply and stop the power supply before performing work.

CAUTION



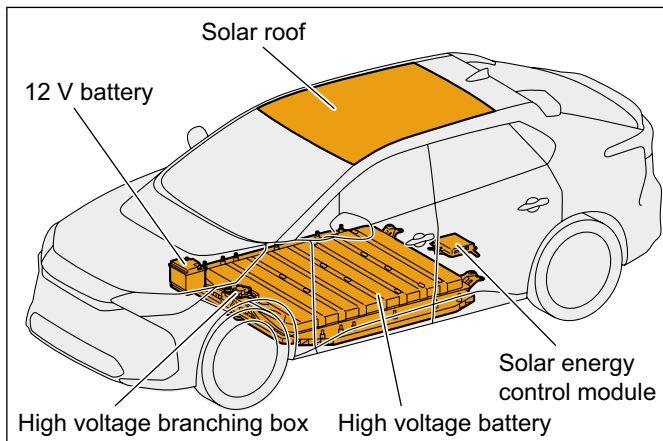
- Electric vehicles have a function for locking the charging cable. The lock can be released by pressing the connector lock switch with the door unlocked.
- If the charging cable lock cannot be released during normal charging, either remove the power supply plug for the external power supply from the outlet, or turn off the main breaker of the external power supply.



- The charging cable cannot be released during quick charging. If charging does not stop even after performing the stop operation for quick charging, turn off the main breaker for the quick charging stand.

■ Solar charging system

- The solar charging system uses the solar roof, which is a photovoltaic cell combined with the roof, to perform photovoltaic electric generation, and charge the high voltage battery while the vehicle is parked depending on the status of the vehicle or compensate for the consumption of the 12 V battery system while driving.
- The solar charging system is comprised of the solar roof, high voltage branching box, solar energy control module, high voltage battery, and 12 V battery.
- An orange high voltage cable is connected between the solar energy control module and the high voltage battery, and a high voltage is applied here while the high voltage battery is charging.
- The solar energy control module is equipped with three DC-DC converters, and it supplies power to the high voltage battery and the auxiliary system.
- The high voltage branching box splits the high voltage wiring between the AC charger and high voltage battery, connecting to the solar energy control module.
- A 15 A fuse is installed for a structure that prevents dangerous phenomena even in the event of a large current.
- The high voltage battery is charged only when the power switch is set to OFF.
- The solar roof can reach a voltage of up to approximately 50 V, depending on conditions such as the sunlight and temperature.
- For the locations where the high voltage devices are equipped on each vehicle, refer to the RS (Rescue Sheet).



WARNING



- Due to the solar charging system, the high voltage system may maintain voltage even while the vehicle is stopped. If the high voltage system is not shut off, the high voltage may cause severe burns or severe injuries due to electrical shocks, and in the worst case, can cause death.

- In order to avoid severe burns or severe injuries or death due to electrical shock, do not touch the orange high voltage cables or high voltage parts. When it is necessary or possible to touch the high voltage parts or high voltage cables, wear appropriate protective equipment such as insulated gloves.
- In cases where a vehicle is stored after responding to the accident or the like and the relevant parties leave the vehicle, there is a risk that other persons may touch the vehicle without knowing and suffer severe injuries or death due to electrical shock. In such a case, display a sign indicating "HIGH VOLTAGE. DO NOT TOUCH DURING OPERATION" to alert other people to the risk. (Make a photocopy of P36 for use.)
- The high voltage battery may be charging even when the vehicle is stopped. To stop charging the high voltage battery, disconnect the ground terminal from the 12 V battery.
- The solar roof generates electricity even from a faint amount of sunlight. To stop generation, cover the entire solar roof with a material that sunlight does not penetrate.

Actions to take for liquid leakage

■ 12 V battery

Dilute sulfuric acid is used in the electrolyte of the 12 V battery.

WARNING



Dilute sulfuric acid is harmful to the human body and may cause inflammation if it is brought into contact with skin or the like. When it is necessary or possible to touch the electrolyte, wear appropriate protective equipment such as rubber gloves and protective goggles when performing work.

CAUTION



The electrolyte includes substances that corrode the coated surface of the vehicle body. If it comes in contact with the vehicle body, there is the risk of changes such as the surface becoming cloudy.

■ High voltage battery



A flammable organic electrolyte composed mainly of carbonic acid ester is used for the electrolyte. The electrolyte is impregnated into the electrode body and separator. In the event that the drive battery (high voltage battery) is broken, the electrolyte may leak out. However, there is no risk of it leaking out in large amounts.

WARNING

- The flammable organic electrolyte composed mainly of carbonic acid ester is harmful to the human body. If you come into contact with the electrolyte or with steam or smoke from a burning battery, it may irritate your eyes, nose, throat, or skin. When it is necessary or possible to touch the electrolyte, wear appropriate protective equipment such as rubber gloves, protective goggles, a protective mask, and a self-contained breathing apparatus (SCBA) when performing work.
- If electrolyte leaked, keep it away from fire and keep good ventilation. Absorb the leaked electrolyte with a waste cloth or the like to collect in an enclosed container.

■ LLC for cooling

The LLC (Long Life Coolant) used for cooling the engine, inverter, and the like includes ethylene glycol, which lowers the freezing temperature, and an anti-corrosion additive for preventing corrosion of metal parts.

■ Lubrication oil

Mineral oil and synthetic oil are used for the oil used as lubrication for the engine, transmission, gears, and the like.

■ Brake fluid

The brake fluid includes several types of glycol ethers as synthetic oil and an anti-corrosion additive for preventing corrosion of metal parts.

CAUTION

The brake fluid includes substances that corrode the coated surface of the vehicle body. If it comes in contact with the vehicle body, there is the risk the coated surface may peel.

■ Power steering fluid

Mineral oil and synthetic oil are used for the power steering fluid.

■ Window washer fluid

Alcohol is included in the window washer fluid to lower the freezing temperature.

Actions to take for gas leakage

■ Refrigerant



- The carbon and fluorine compound R-134a or R-1234yf is used for the refrigerant gas for the air conditioning.
- It is colorless, odorless, and harmless to the human body.

■ Nitrogen (N₂) gas



- Nitrogen (N₂) is used for the gas filled damper.
- It is colorless, odorless, and harmless to the human body.

Wearing of protective equipment during high voltage work

- Insulated gloves
- Rubber gloves (used for processing electrolyte)
- Protective goggles (eye protection goggles)
- Rubber boots for electrical purposes
- Solvent resistant apron (for organic solvents)
- Protective mask (for organic gases)

CAUTION



If you touched electrolyte, follow the guideline below.

■ If electrolyte contacts your skin

- If electrolyte contacts your skin directly, immediately wash it off completely using plenty of water with soap or a skin cleaner.
- Be careful to avoid rubbing your eyes using gloves or the like that have been in contact with electrolyte.
- In case of an injury or a sharp pain, consult a medical specialist.

■ If electrolyte enters your eyes

If electrolyte enters your eyes, loudly call for help. Do not rub your eyes. Immediately wash the liquid away even from the inside of the eyelid with plenty of water for 15 minutes or more, and then consult a medical specialist.

■ **If someone swallows electrolyte by mistake**

- Do not force that person to throw up.
- Give the concerned party plenty of water to drink to dilute the electrolyte.
- Do not give the person any water to drink if they are unconscious.
- If vomiting occurs naturally, prevent the person from suffocating.
- Take the person to the nearest emergency medical institution.

■ **If you inhale the vapor of electrolyte**

- Immediately blow your nose and gargle your throat.
- Carry the person to a safe location and give them oxygen.
- Take the person to the nearest emergency medical institution.

6. In case of fire

When performing initial fire extinguishing activities, use a large amount of water from a fire hydrant. This will also cool down the vehicle.



WARNING



When a vehicle fire occurs, gas that is harmful to the human body is generated from plastics and the like used in the vehicle. When extinguishing fires, wear appropriate protective equipment such as a mask when performing work.

■ Fire extinguishing agent

- Water is an appropriate fire extinguishing agent.
- If using a fire extinguisher to extinguish fires, use a fire extinguisher that is effective not only for normal fires (general fires where solid objects are burning), but also oil fires (fires involving gasoline, petroleum, oil, and the like) and electrical fires (fires involving electrical wiring, electrical devices, and the like).

■ Vehicles equipped with a high voltage battery



- Hybrid electric vehicles (HEV) and electric vehicles (EV) are equipped with a high voltage battery.
- In order to cool off the battery, extinguish fires using plenty of water.
- For information about the high voltage battery location on each vehicle, refer to the RS (Rescue Sheet) for each vehicle.
- Extinguish battery fires from upwind to avoid smoke and flames.
- Use a large amount of water for fire extinguishing, in order to cool the battery.

Spraying water directly on the high voltage battery increases the cooling effect. Open the door or break the window when extinguishing the fire.

- Damaged cells of the high voltage battery may experience thermal runaway or reignition.

To prevent thermal runaway and reignition, responders must be aware of the risk of remaining energy. The main cause of thermal runaway is a short circuit inside the high voltage battery cell and the resulting increase in the cell's internal temperature.

If thermal runaway occurs in one cell, it may spread to many cells.

- After an accident, energy remains inside the undamaged high voltage battery cells.

Due to the remaining energy, the high voltage battery may reignite multiple times even after the fire is extinguished.

- The flammable organic electrolyte composed mainly of carbonic acid ester is harmful to the human body. If you come into contact with the electrolyte or with steam or smoke from a burning battery, it may irritate your eyes, nose, throat, or skin. When it is necessary or possible to touch the electrolyte, wear appropriate protective equipment such as rubber gloves, protective goggles, a protective mask, and a self-contained breathing apparatus (SCBA) when performing work.

WARNING



- In order to avoid severe burns or severe injuries or death due to electrical shock, do not touch the orange high voltage cables or high voltage parts (such as the high voltage battery). When it is necessary or possible to touch them, make sure to wear insulated gloves and insulate the orange high voltage cables and high voltage parts using insulating tape.
- If you perform fire extinguishing using a small amount of water, there is a possibility a short circuit may occur inside the high voltage battery, causing the fire to restart.

CAUTION



- If it is deemed difficult to spray a large amount of water at the high voltage battery, we recommend allowing the high voltage battery to burn out.
- If you notice leaking fluid, sparks, smoke, unusual noise, or the like near the high voltage battery, there is a risk of a battery fire.

7. In case of submersion

Pull the vehicle out of the water, secure the vehicle (P7), stop the vehicle (P11), and then perform rescue work.

CAUTION



- If a vehicle has been submerged, there is the risk of short circuits occurring due to electrolytic corrosion (that is, corrosion of the wiring and substrates due to an electrochemical action caused by water), which may cause a vehicle fire after time has passed.
- As there is a risk of a vehicle fire, absolutely never turn the ignition switch or power switch of a vehicle that has been submerged to ACC or ON.

■ Vehicles equipped with a high voltage battery



- If a vehicle equipped with a high voltage battery was partially or completely submerged, as the vehicle body and high voltage circuit are insulated, touching the vehicle body does not pose the risk of an electrical shock.
- As the water and the vehicle are isoelectric, entering the water does not pose the risk of an electrical shock.

WARNING



- Touching any exposed orange high voltage cables or high voltage devices including the high voltage battery can create electric potential, resulting in an electrical shock.
- In order to avoid severe burns or severe injuries or death due to electrical shock, do not touch the orange high voltage cables or high voltage parts (such as the high voltage battery). When it is necessary or possible to touch them, make sure to wear insulated gloves and insulate the orange high voltage cables and high voltage parts using insulating tape.

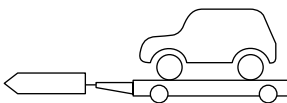
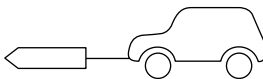
8. Towing / Transportation / Storage

Towing

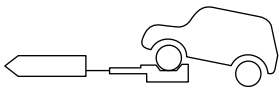
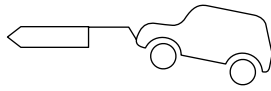
WARNING



- When transporting the vehicle, wear insulated gloves and detach the service disconnect plug before transportation.
- In order to avoid severe burns or severe injuries or death due to electrical shock, do not touch the orange high voltage cables or high voltage parts (such as the high voltage battery). When it is necessary or possible to touch them, make sure to wear insulated gloves and insulate the orange high voltage cables and high voltage parts using insulating tape.

Towing method	Precautions	Except for SHEV		SHEV
		AWD	FWD	
Lifting up four wheels (on a trailer)  NT-00023	Towing the vehicle after lifting up all four wheels is a basic rule for an AWD model.	○	○	○
Rope  NT-00024	• Check if both front and rear wheels are rotated normally. • Driving conditions for a CVT model: Driving speed of 30 km/h (19 MPH) or less. Driving distance of 30 km (19 miles) or less.	▲	▲	×

Marked ○ : OK, Marked × : Prohibited, Marked ▲ : Conditionally OK

Towing method	Precautions	Except for SHEV		SHEV
		AWD	FWD	
Raising the front wheels  NT-00025	This is prohibited for the AWD model.	×	○	×
Lifting up the front wheels  NT-00026	• Prohibited, due to damage on bumper, front grille, etc. • Do not raise the vehicle using the bumper.	×	×	×

Marked ○ : OK, Marked × : Prohibited, Marked ▲ : Conditionally OK

NOTE

For information on how to tow each vehicle, refer to the RS (Rescue Sheet) for each vehicle.

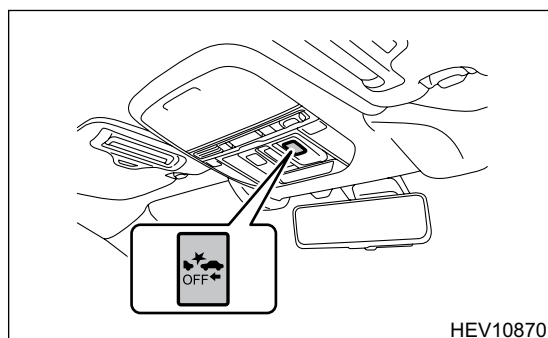
CAUTION



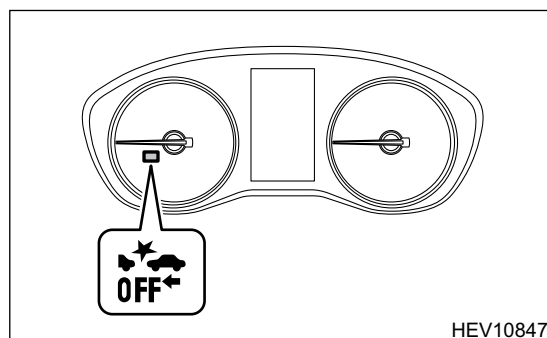
- If the 12 V battery has run out, or if the ground terminal of the 12 V battery has been cut off, you may not be able to shift the lever out of the P range.
- If there is a malfunction in the transmission interior or drive system, lift up four wheels (on a trailer) for towing.
- If there is a long or steep downhill road along the way, lift up four wheels (on a trailer) for towing. As the engine brake does not operate at all while the vehicle is being towed with a rope, continuously stepping on the brake pedal while going downhill can cause the brake to overheat, which will prevent the brake from operating.
- Do not lift up the rear wheels to avoid unsteady rotation.
- Do not use the towing hook (eyebolt) except when towing.
- Make sure to detach the towing hook (eyebolt) after towing. If it remains attached, the airbag and the autodisconnect may not operate properly when receiving a shock. It may also affect the collision performance of the vehicle.

- Be careful of the following items when towing a vehicle with a rope.
 - Towing may not be possible depending on the content of the malfunction.
 - Turn the power to "ACC" or "ON" and check that the steering wheel moves freely.
 - When towing, turn the power ON and the engine OFF (do not allow the hybrid system to operate).
 - When towing, place the shift lever in the "N" position.
 - Release the parking brake to avoid tire dragging.
 - When towing with the power ON and the engine OFF (do not allow the hybrid system to operate) and not in the P range, the "pedestrian alert" may sound. However, this is not a malfunction.
 - Drive so as not to apply an impact on the rope, such as by suddenly starting.
 - As the power steering does not work, be aware that the steering wheel operation will feel heavy. (When engine is stopped)
 - As the servo brake does not work, be aware that the brake will not be applied effectively. (When engine is stopped)
 - When towing a vehicle equipped with Subaru EyeSight, be sure to hold down the pre-collision brake OFF switch to turn OFF the pre-collision brake function. (Check that the pre-collision brake OFF indicator light in the meter lights up.)

▼Pre-collision brake OFF switch

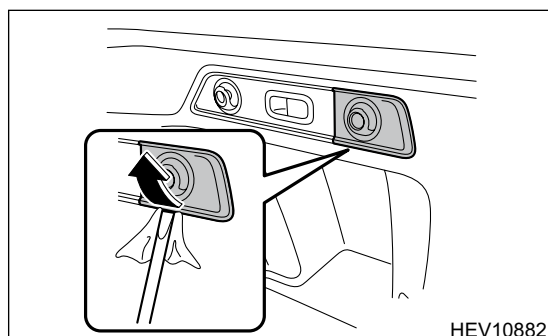


▼Pre-collision brake OFF indicator light

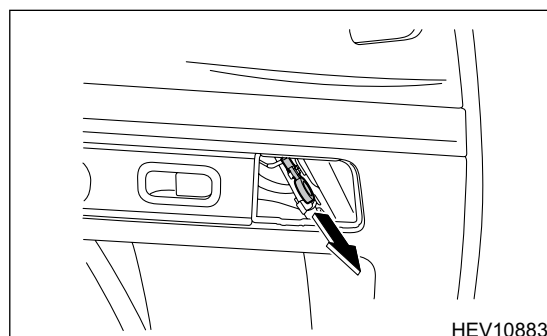


- To release the fuel lid, press (Unlock) on the driver's center door lock switch, and the access key (Unlock) button, or set the push button ignition switch to ACC or OFF. If the fuel lid cannot be released, perform the following operation (electric fuel lock).
 - Use a screwdriver to remove the shopping bag hook panel on the right side of the cargo area. Remove the lever, and then lightly pull towards the vehicle rear (by about 1 cm) to release.

▼Hook panel



▼Lever



Method for manually releasing the electronic parking brake

CAUTION



- Only do this when releasing the electronic parking brake in an emergency.
- If you manually released it, replace the parking actuator.

1. Park the vehicle in a safe location and secure the front tires with wheel stoppers.
2. Remove the ground terminal from the 12 V battery.

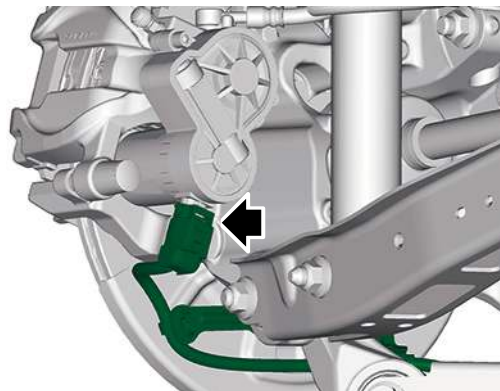
CAUTION



If mud, water, or the like is present around the connector, clean that off before removal.

3. Jack up the rear side of the vehicle and support it with rigid racks or the like.
4. Remove the rear wheel on the unreleased side.
5. Disconnect the connector from the parking brake actuator.

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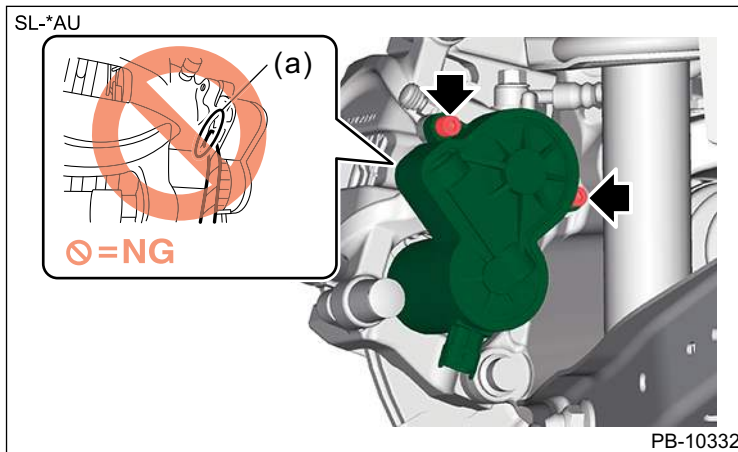
PB-10324

6. Using a hexagon wrench, remove the bolts from the parking brake.

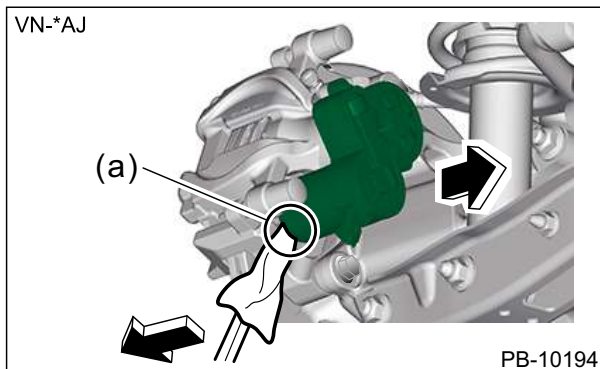
CAUTION



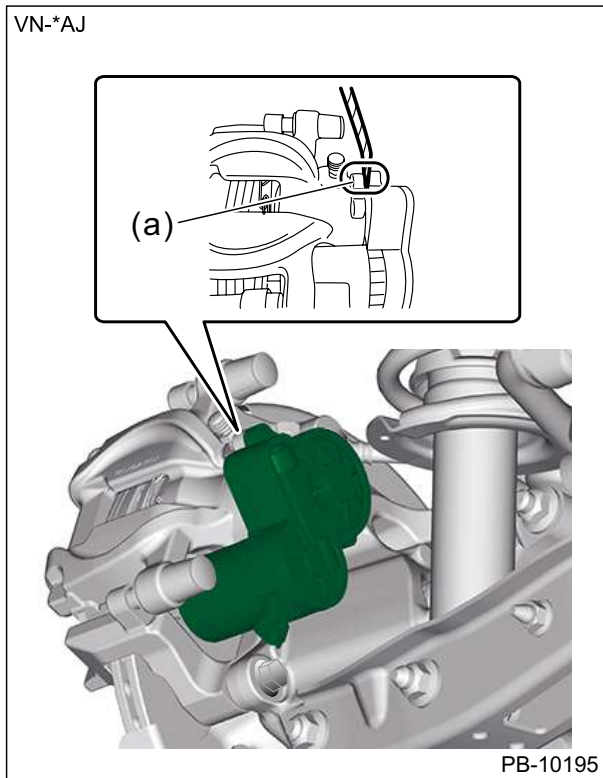
- The parking brake actuator turns due to reaction force generated by the motor. Be careful not to pinch your hand.
- Do not insert a tool into the seal surface (a) on the parking brake actuator.



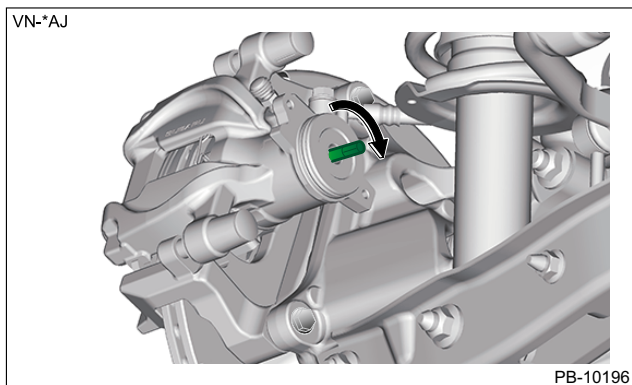
7. Insert a screwdriver or the like into section (a) and press the parking brake actuator out toward you.



8. Insert a screwdriver or the like into the gap made by pushing the parking brake actuator and remove it.



9. Using TORX® E12, turn the spindle bolt clockwise approx. 1 rotation.



10. Confirm that the parking brake lock is released and the disc rotor rotates.

Storing the damaged vehicle

After removing the gasoline, oil, and the like, store the vehicle with the ground terminal of the 12 V battery removed.

■ Submerged vehicle

In addition to the normal actions, store the vehicle with the water drained.

CAUTION



- If a vehicle has been submerged, there is the risk of short circuits occurring due to electrolytic corrosion (that is, corrosion of the wiring and substrates due to an electrochemical action caused by water), which may cause a vehicle fire after time has passed. When storing a vehicle that has been submerged, do so in a well-ventilated location 15 m or more away from other things.
- As there is a risk of a vehicle fire, absolutely never turn the ignition switch or power switch of a vehicle that has been submerged to ACC or ON.

■ Vehicles equipped with a high voltage battery

In addition to the normal actions, store the vehicle with the service disconnect plug removed.

WARNING



- The service disconnect plug is a high voltage part. Touching it without protective equipment is dangerous and can result in severe burns or severe injuries or death due to electrical shock. If touching the service disconnect plug, wear appropriate protective equipment such as insulated gloves.
- Even after disabling the vehicle (refer to P11) and removing the service disconnect plug, high voltage is stored in the high voltage battery unit.
- In order to avoid severe burns or severe injuries or death due to electrical shock, do not touch the orange high voltage cables or high voltage parts (such as the high voltage battery). When it is necessary or possible to touch them, make sure to wear insulated gloves and insulate the orange high voltage cables and high voltage parts using insulating tape.
- In cases where a vehicle is stored after responding to the accident or the like and the relevant parties leave the vehicle, there is a risk that other persons may touch the vehicle without knowing and suffer severe injuries or death due to electrical shock. In such a case, display a sign indicating "HIGH VOLTAGE. DO NOT TOUCH DURING OPERATION" to alert other people to the risk. (Make a photocopy of P36 for use.)

CAUTION

The high voltage battery poses the risk of a short circuit occurring inside the battery due to the impact from the accident or electrolytic corrosion, which may cause a vehicle fire after time has passed. When storing a vehicle equipped with a high voltage battery, do so in a well-ventilated location 15 m or more away from other things.

9. Important additional information

Gas filled dampers

- Gas filled dampers are used in various locations in the vehicle, such as in the suspension or in the hood, and for various purposes.
- Nitrogen (N₂) gas is colorless, odorless, and harmless to the human body.
- For information about the location where the gas filled dampers are equipped on each vehicle, refer to the RS (Rescue Sheet) for each vehicle.

WARNING



- If a gas filled damper is heated when a vehicle fire occurs, the sealed nitrogen (N₂) gas may expand, rupturing the damper and causing parts to fly off, resulting in unexpected injuries.
- If a gas filled damper is cut, the sealed nitrogen (N₂) gas may cause cutting powder and the like to fly. When it is necessary to cut it, wear appropriate protective equipment such as protective goggles when performing work.

■ Front and rear suspension dampers

Suspension dampers are equipped on the front and rear suspensions.

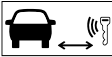
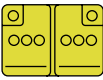
■ Engine hood (Bonnet) dampers

Engine hood (bonnet) dampers are equipped to the support sections for the engine hood (bonnet).

■ Trunk / rear gate dampers

Each damper is equipped to the support sections for the trunk or rear gate.

10. Explanation of pictograms used

	Air-conditioning component		Bonnet		Boot
	Steering wheel, tilt control		Seat height adjustment		Seat adjustment, longitudinal
	Seat adjustment, seat back recline		General warning sign		Warning, Electricity
	Remove smart key		Use thermal Infrared camera		Explosive
	Flammable		Gases under pressure		Corrosives
	Hazardous to the human health		Acute toxicity		Use ABC powder to extinguish the fire
	Use water to extinguish the fire		Lifting point; central support		Fuse box disabling high voltage
	Battery low voltage		High voltage device that disconnects high voltage		High voltage power cable